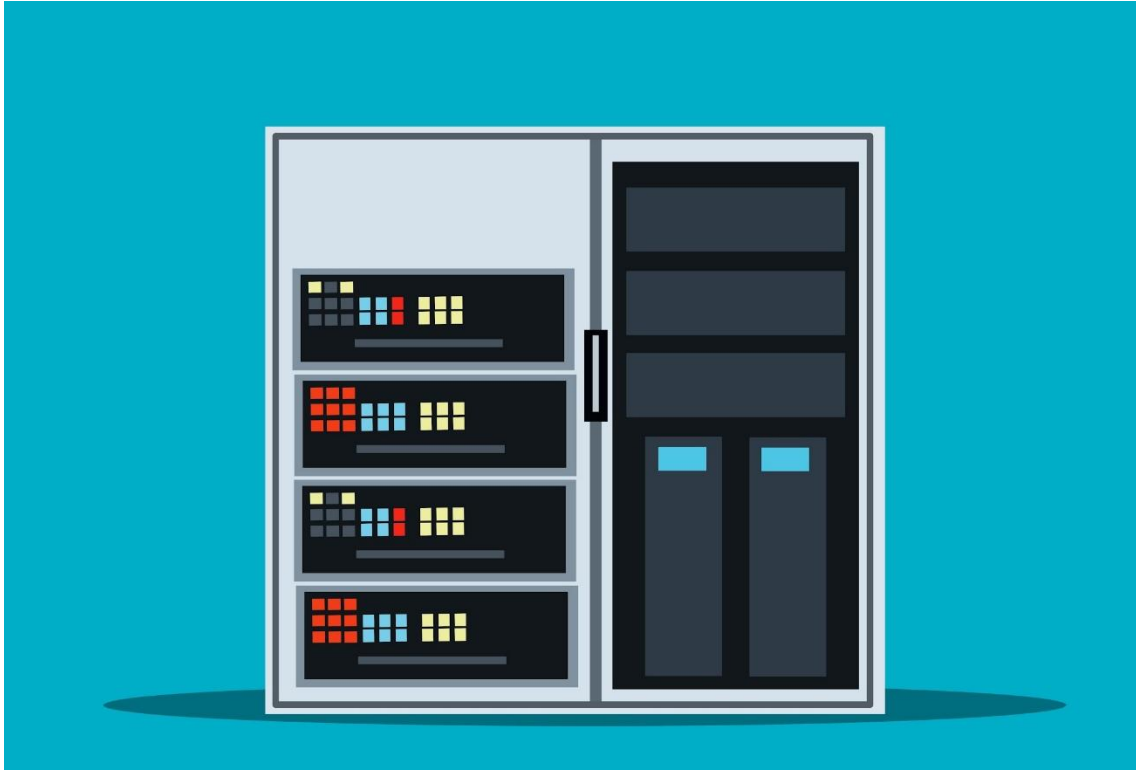


Backup, Recover, & Turnover Management

By Pratyush Bishowkarma



This document outlines a comprehensive **Backup and Recovery Plan** for our corporate database environment, which supports approximately 1,500 employees and generates \$250 million in net sales, anticipated to grow by 10% annually. As the lead DBA of a team of three, the plan will cover five key sections.

First, it will detail backup procedures for our MS SQL Server database, including legacy systems like accounting and forecasting, as well as our cloud-based HR system. This section will address scheduling retention periods, and security measures.

Second, support protocols will be established to manage unexpected events, including staff turnover. Third, the recovery processes for local and cloud systems will be defined. Fourth, we will create metrics to evaluate and improve our backup and recovery strategies. Finally, a turnover procedure will ensure proper documentation and knowledge transfer within the DBA team.

Backup Models

Backup Models – Robust

Robust Model	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
12 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
1 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
2 AM	Full	Full	Full	Full	Full	Full	Full
3 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
4 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
5 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
6 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
7 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
8 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
9 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
10 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
11 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
12 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
1 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
2 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
3 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
4 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
5 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
6 PM	Full	Full	Full	Full	Full	Full	Full
7 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
8 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
9 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
10 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
11 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction

This model is extremely extensive with multiple full log backups and multiple transaction log backups to ensure proper server runtime.

Backup Model – Intermediate

Robust Model	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
12 AM							
1 AM							
2 AM	Full	Full	Full	Full	Full	Full	Full
3 AM							
4 AM							
5 AM							
6 AM							
7 AM							
8 AM	Full	Full	Full	Full	Full	Full	Full
9 AM							
10 AM							
11 AM							
12 PM							
1 PM							
2 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
3 PM							
4 PM							
5 PM							
6 PM							
7 PM							
8 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
9 PM							
10 PM							
11 PM							

This model is basic with only two transaction log backups a day and two full backups per day.

Backup Model – Simple

Robust Model	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
12 AM							
1 AM							
2 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
3 AM							
4 AM							
5 AM							
6 AM							
7 AM							
8 AM							
9 AM							
10 AM							
11 AM							
12 PM							
1 PM							
2 PM	Full						
3 PM							
4 PM							
5 PM							
6 PM							
7 PM							
8 PM							
9 PM							
10 PM							
11 PM							

The simple model is what DBA would use as it requires a full backup up only once a week and only one transaction log backup per day.

Support Protocols

The purpose of this support protocol is to define how database incidents are reported, escalated, and resolved across both on-site SQL Server environment and the cloud-based system. This protocol ensures business continuity, clear accountability, and no single point of failure in staffing or communication.

Both environments follow the same technical standards, but ownership and escalation paths differ depending on whether the system is local or cloud hosted.

System Prioritization

All systems are assigned to a priority level based on business impacts. This determines response time, escalation speed, and staffing involvement.

Priority 1 – Mission Critical

Includes:

- Order Entry
- Accounting Database

These systems directly impact revenue and financial operations. Any outage here affects business continuity immediately. Target time is 15 minutes. DBA engagement is mandatory.

Priority 2 – High Importance

Includes:

- Fulfillment systems
- Cloud-based HR SQL Server

These system impact operations and employee management. Outage is serious but not immediately revenue-blocking. Target response time is 30 minutes.

Priority 3 – Moderate Impact

Includes:

- Planning and forecasting databases

These systems support internal analysis. Downtime is tolerable for short periods. Target response time is 1 hour.

Priority 4 – Low Impact

Includes:

- Data warehouse (reporting only)

These systems support analytics. Short outages do not stop core operations. Target response time is 2 hours.

This prioritization ensures resources are focused where business risk is highest.

Contact Information and Escalation Method

All database incidents must be initiated through the IT Helpdesk. This creates a formal ticket record for tracking and audit purposes.

Helpdesk Responsibilities:

- Log incident in ticketing system
- Gather error details and screenshots
- Identify affected users and systems
- Escalate database-related issues to DBA team

Contact methods includes:

- IT Service Portal
- Support email
- Internal support hotline

DBA Structure:

- Primary DBA leads incident response
- Secondary DBA provides support to Primary DBA and has full power when Primary DBA is unavailable
- Tertiary DBA provides additional support and monitoring

If the issue is not resolved within defined Service Level Agreement, then;

- IT Director is notified
- CIO is notified for extended outages
- If cloud-hosted infrastructure is involved, vendor enterprise support is engaged.

Staffing Backup and Continuity

To prevent operational dependency on one individual, the organization maintains structured staffing redundancy.

- Three DBAs are cross trained on all backup and recovery procedures
- All runbooks and recovery documentation are centrally stored
- Credentials are stored in an enterprise password vault
- Quarterly knowledge transfer sessions are conducted

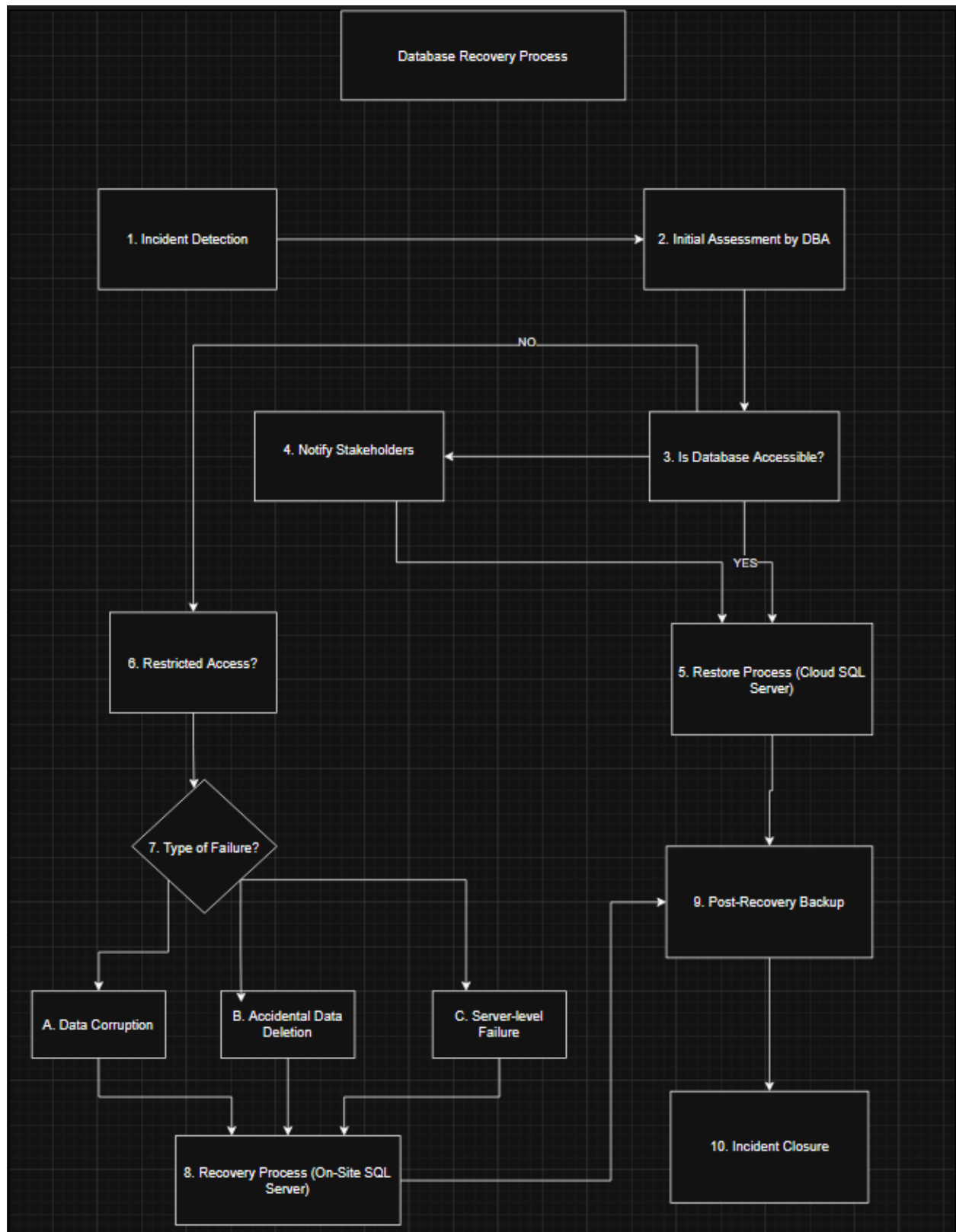
In the event of planned turnover:

- Knowledge transfer must be completed before final workday
- Access rights are reviewed and revoked
- Service account passwords are rotated

In the event of unexpected absence:

- Secondary DBA automatically assumes Primary responsibilities
- No executive approval required to initiate recovery
- All procedures are documented to allow immediate continuation of operations

Recovery Process



Metrics

1. Backup Job Success Rate – Measures the percentage of scheduled backups jobs that complete successfully without errors during a given reporting period.
2. Restored Testa Success Rate - Measures the percentage of test restorations that complete successfully and result in a fully functional database.
3. Average Full Backup Duration – Tracks the average time required to complete a full database backup.
4. Average Restore Duration – Measure the average time required to fully recover a database from backups.
5. Recovery Time Objective (RTO) Compliance – Compares the actual recovery time to the defined Recovery Time Objective to ensure recovery meets business expectations.
6. Recovery Point Objective (RPO) Compliance – Compares actual data loss time to the defines Recovery Point Objective to verify acceptable data loss limits are maintained.
7. Backups Failure Frequency – Measures how often backup jobs fail within a defined time period, such as monthly.
8. Database Storage Growth Rate – Tracks the percentage increase in database size over time to support capacity planning.
9. Total System Downtime per Incident – Measures the total duration that a database system remains unavailable during an outage.
10. Number of Users Impacted per Outage – Records how many users were affected during each database incident to evaluate business impacts.

Turnover Process

The organization recognizes three backup modules that may be used depending on system criticality, risk tolerance, and business impact. The Robust Backup Module is designed for mission-critical systems such as Order Entry and Accounting, where downtime or data loss would significantly impact revenue. In this model, transaction log backups are performed hourly throughout the day, with full backups performed twice daily at 2 AM and 6 PM. This approach provides very tight Recovery Point Objective and minimizes potential data loss to a very small window. It also increases storage usage and administrative monitoring, but it offers the highest level of protection and is appropriate for high-availability environments.

The Intermediate Backup Model is used for system with moderate business impact. In this structure, full backups are performed twice daily at 2 AM and 8 AM, and transaction log backups are performed twice daily at 2 PM and 8 PM. This model reduces storage and processing overhead compared to the robust model while still maintaining reasonable recovery capabilities. It provides a balanced approach between operational efficiency and data protection, making it suitable for planning, forecasting, or internal operational systems where short data gaps are acceptable.

The Simple Backup Model is intended for low-risk systems such as reporting databases or development environments. In this model, a full backup is taken once per week, and a single transaction log backup is taken daily at 2 AM. This approach minimizes administrative workload and storage requirements but allows for a larger potential data loss window. It is not recommended for revenue-impacting systems but may be sufficient for non-critical workloads. Selection of the backup model is determined by system priority classification and approved by the DBA team as part of overall backup governance.

CHANGE REQUEST FORM

CHANGE STATEMENT	Provide a brief description of the requested change	CHANGE No.	
		REPORT ID	
		DATE	
LOCATION		TIME	

REQUESTED BY	TITLE	CONTACT INFORMATION

REQUEST PRIORITY	LOW	MEDIUM	HIGH	CHANGE TYPE	MINOR	MAJOR	STANDARD	EMERGENCY

DESCRIPTION OF CHANGE	Provide a detailed description of the problem and circumstances leading to the requested change

SUPPORTING INFORMATION	Provide screenshots of an error or attach the error in a document / report

REASONS AND JUSTIFICATION	Describe the reason why the change has been requested and the justification for the request

AFFECTED AREAS According to the perception of the requester	System	
	Subsystem	
	Documentation	

DECISION		Project Manager Name	Signature	Date
	ACCEPTED			
	REJECTED	Approver Name & Title	Signature	Date
	More Info Requested			

Source: <https://www.smartsheet.com/content/change-request-form-templates>

ID & Kill User

SQLQuery5.s... \praty (54)* SQLQuery4.sq... \praty (52)*

1

sp_who

400 % No issues found

	spid	ecid	status	loginame	hostname	blk	dbname	cmd	request_id
32	33	0	background	sa		0	master	UCS_TASK_NO_SPEC	0
33	34	0	sleeping	sa		0	master	TASK MANAGER	0
34	35	0	background	sa		0	master	UCS_TASK_NO_SPEC	0
35	36	0	background	sa		0	master	UCS_TASK_NO_SPEC	0
36	39	0	background	sa		0	NULL	XE DISPATCHER	0
37	40	0	background	sa		0	master	TRACE QUEUE TASK	0
38	41	0	background	sa		0	NULL	RECEIVE	0
39	42	0	background	sa		0	master	TASK MANAGER	0
40	43	0	background	sa		0	master	CHECKPOINT	0
41	44	0	background	sa		0	master	BRKR EVENT HNDLR	0
42	45	0	background	sa		0	master	BRKR TASK	0
43	46	0	background	sa		0	master	BRKR TASK	0
44	47	0	background	sa		0	master	BRKR TASK	0
45	50	0	sleeping	sa		0	master	TASK MANAGER	0
46	51	0	sleeping	NT SE...	LOG	0	master	AWAITING COMMAND	0
47	52	0	sleeping	log\praty	LOG	0	Sick	AWAITING COMMAND	0
48	53	0	sleeping	log\praty	LOG	0	master	AWAITING COMMAND	0
49	54	0	runnable	log\praty	LOG	0	Sick	SELECT	0
50	59	0	sleeping	sa		0	master	TASK MANAGER	0
51	60	0	sleeping	sa		0	master	TASK MANAGER	0
52	66	0	sleeping	sa		0	master	TASK MANAGER	0
53	74	0	background	sa		0	NULL	PARALLEL REDO TASK	0
54	80	0	sleeping	sa		0	master	TASK MANAGER	0
55	82	0	sleeping	sa		0	master	TASK MANAGER	0
56	86	0	sleeping	log\praty	LOG	0	master	AWAITING COMMAND	0
57	88	0	sleeping	sa		0	master	TASK MANAGER	0
58	94	0	background	sa		0	NULL	PARALLEL REDO TASK	0

1 sp_who

2

3 kill 52

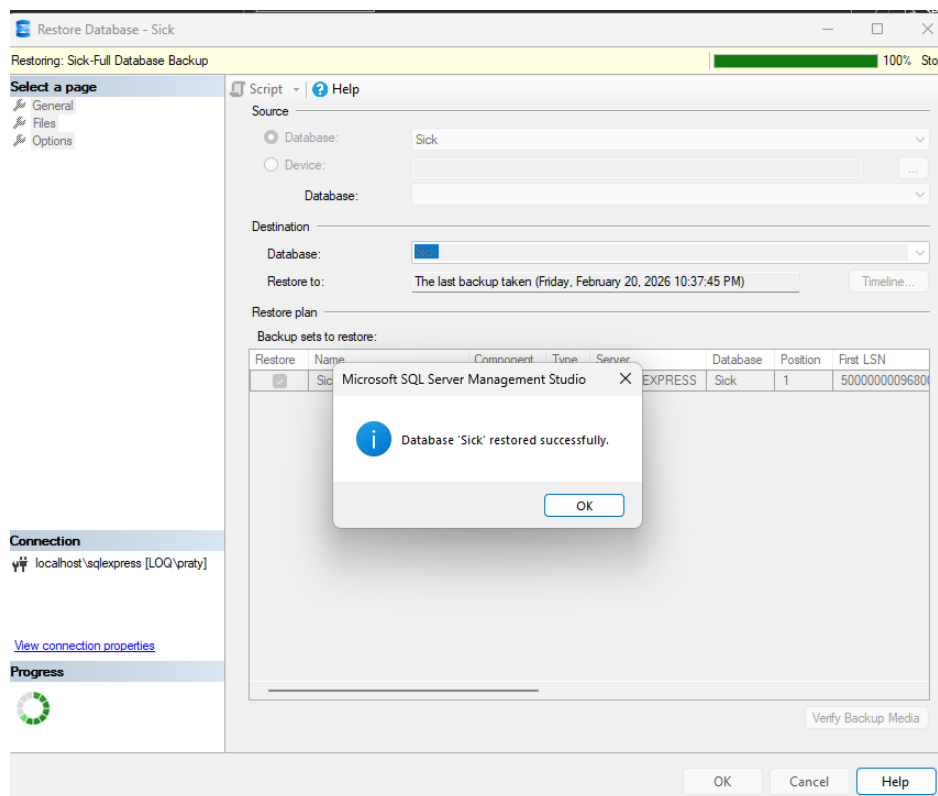
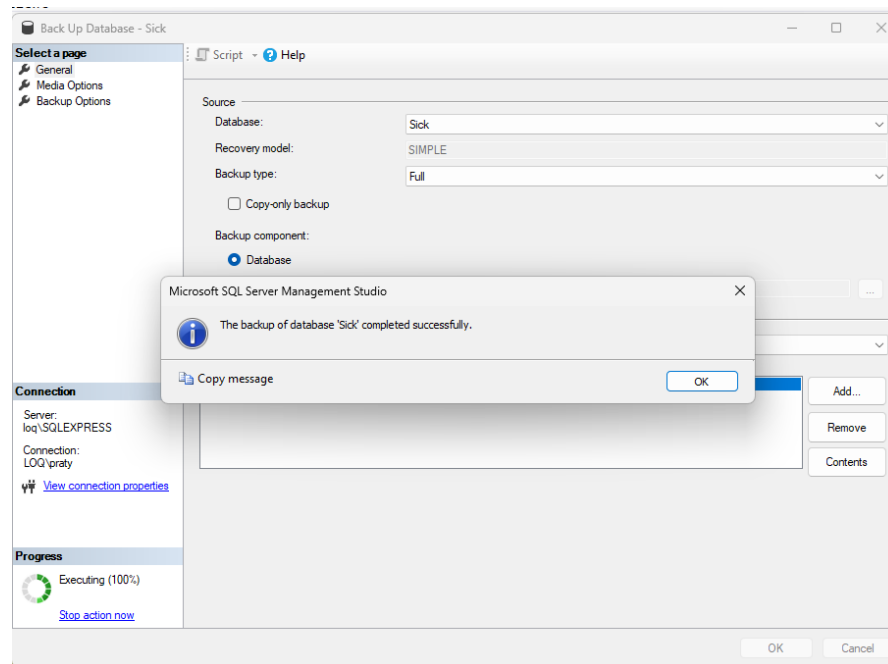
200 % No issues found

Messages

Commands completed successfully.

Completion time: 2026-02-20T23:09:48.3851262-05:00

Backups/Recovery with Wizard



Backups/Recovery with Scripts

```
1  BACKUP DATABASE sick
2  TO DISK = 'C:\Backups\sick.bak'
3  WITH STATS = 10;
4
5  PRINT 'Backup completed successfully.';
6
```

61% ▾ ● 1 ▲ 0 ↑ ↓

Messages

```
10 percent processed.
20 percent processed.
30 percent processed.
40 percent processed.
50 percent processed.
60 percent processed.
70 percent processed.
80 percent processed.
90 percent processed.
100 percent processed.
Processed 856 pages for database 'sick', file 'sick' on file 2.
Processed 2 pages for database 'sick', file 'sick_log' on file 2.
BACKUP DATABASE successfully processed 858 pages in 0.115 seconds (58.254 MB/sec).
Backup completed successfully.
```